Name & Surname **Polimi ID**

Exercise 1

A. Please describe the parallel gather pattern, including assumptions, complexity, and examples. (1)

B. Please describe how the Split parallel pattern works. How is it implemented (pseudo code required)? (2)

C. How and under what conditions can the following code be parallelized? Is there a way to write the code for better parallelization? (1)

```
for (i=10; i<100; i++)
        for (i=10; i<100; i++)
           a[0][j] += f(a[i][j-10]);
```

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Exercise 2

A. How can a CUDA programmer judge if an access is coalesced? (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

B. Describe the corner turning technique. How the size of the shared memory is defined in this technique? (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**

C. Please describe the sequential consistency memory model. (1)

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Exercise 3

A. Given the following CUDA kernel, discuss the impact of control divergence for different N values. (3)

```
__global__ void myKernel(int *data, int N) {
     int idx = threadIdx.x + blockIdx.x * blockDim.x;
     if (idx < N) {
          if (idx % 2 == 0) {
               data[idx] *= 2;
          } else {
               data[idx] += 1;
          }
     }
}
```

B. Describe the types of memory available on NVIDIA GPUs and who can access each of them. (1)

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Exercise 4

A. Why is heterogeneous processing becoming increasingly relevant in high-performance computing? (1)

B. Describe the trade-offs that can be explored in the design space created by the Halide scheduling directives. (2)

C. Provide at least two examples of situations where a combination of different parallel programming languages is used. (1)